

**APPLICATION OF THE LAGRANGIAN FORMULATION IN
RAY PROPAGATION IN A MEDIUM WITH A
SPHERICALLY SYMMETRIC INDEX OF REFRACTION
PROFILE**

by

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An Undergraduate Thesis Submitted to the
National Institute of Physics
College of Science
University of the Philippines
Diliman, Quezon City

In Partial Fulfillment of the Requirements
for the Degree of
Bachelor of Science in Physics

April 2012

CERTIFICATION

This is to certify that this undergraduate thesis entitled “**Application of the Lagrangian Formulation in Ray Propagation in a medium with a Spherically Symmetric Index of Refraction Profile**” submitted by **Cye Tamura** to fulfill part of the requirements for the degree of **Bachelor of Science in Physics** was successfully defended and approved on 24 March 2012.

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Acknowledgments

Keeping it short:

Firstly, I would like to thank Sir Perry for being an awesome adviser and guiding me through the making of this thesis.

I would also like to thank my thesis readers, Ma'am Pastor and Sir Alis, for being very helpful and for giving very constructive suggestions and comments.

And for helping me with my programming, thank you, Myles and Kendrick. Also, thanks to Diane for assisting me with the thesis formats and other miscellaneous.

ABSTRACT

APPLICATION OF THE LAGRANGIAN FORMULATION IN RAY PROPAGATION IN A MEDIUM WITH A SPHERICALLY SYMMETRIC INDEX OF REFRACTION PROFILE

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Spherically symmetric graded refractive index profiles are prevalent in studies of optical and acoustic behaviour in the ocean and the atmosphere. We will be considering a specific refractive index profile in which the speed of the ray increases as the distance from a certain radius increases. This will be described by piecewise functions. An analogy between rays and particles are used to solve the systems, and ray trajectories within these media are plotted.

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Chapter 1

Introduction

1.1 Background of the Study

There are many cases of waves such as sound and light traversing an inhomogeneous medium with a spatially varying, graded index of refraction. This usually results in the ray travelling along a curved path.[1]

For light, we can find graded-index profiles when there is a temperature gradient in the atmosphere. The rarefaction of air in the atmosphere due to altitude is another example. [2] Graded-index lenses are also another relevant area of study. [3]

For sound, we can find many examples of graded-index profiles in the ocean and the atmosphere. Differences in temperature between air masses cause a curvature in the propagation of the sound rays. [4]

1.2 The RHO Profile

In this paper, we will consider a specific spherically symmetric refractive index profile. It is described by:

$$n^2(r) = \begin{cases} n_b + C_1 - C_2 r^2 & , r_0 < r < b \\ n_a + C_3 r^2 & , a < r \leq r_0 \\ n_a & , r > \sqrt{\frac{C_1}{C_2}} \end{cases} \quad (1.1)$$

Where a and b are the boundaries of the medium and r_0 is where the shift in refractive indices occurs. n_a and n_b are minimum squared indices of refraction, to prevent the index from becoming ≤ 0 . n_a and n_b is are refractive indices beyond the inner and outer boundaries of the medium. We also set $C_1 = C_3 r_0^2 + n_b - n_a$ to make a continuous refractive index profile.

This profile was chosen because the trajectories for the two index profiles can be determined analytically by exploiting an analogy between rays and particles. This profile is referred to as the radial harmonic oscillator profile or RHO profile because its trajectories follow those of a particle under RHO and inverse RHO potentials. This will be discussed in Chapter 2.

A piecewise refractive index was constructed out of these two profiles in order to create a function with an increasing $n(r)$ until it reaches its maximum at r_0 , where it decreases until it reaches a constant. Figure 1.1 shows a graphical representation of the index of the RHO profile.

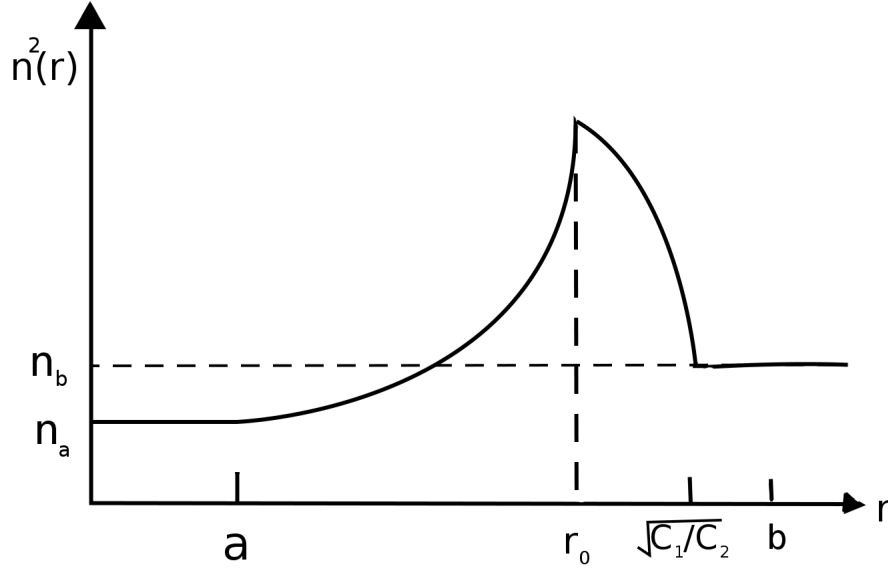


Figure 1.1: A plot showing $n^2(r)$ vs. r for this refractive index profile

In chapter 2, we will use the analogy between rays and particles to derive equations of motion for a ray propagating in our refractive index profile. In chapter 3, we will plot ray trajectories for this profile. We will also see how varying several parameters influence the ray trajectories. In chapter 4, the conclusion and recommendations will be given. Appendix A will show preliminary results of how a ray pulse emitted inside the medium evolves over time. Appendix B uses the same Lagrangian formulation to determine ray trajectories for a common index of refraction profile. Real world values were used as parameters and the results were compared with the properties of sound rays propagating in SOFAR channels.

1.3 Occurrence of the Refractive Index Profile

Some cases of spherically symmetric refractive index profiles that gradually increase up to a certain radius then begin to decrease can be found in the ocean and atmosphere.

In the ocean, The speed of sound changes as a function of temperature, salinity, and pressure. [5] The change in temperature near the surface of the ocean is small, until it reaches a layer called the thermocline. Here, the temperature drops much faster with depth. Beyond the thermocline, the changes in temperature are small. The speed of sound in the ocean is greatly affected by this temperature change while it is closer to the surface. However, as the depth in the ocean increases, the water pressure becomes a more dominant factor in influencing the speed of sound. Thus, the speed of sound decreases with depth until reaching a minimum value. Beyond this depth, the speed of sound begins increasing again.

This has an interesting effect on sound propagation near the thermocline. As sonic pulses approach the surface they curve back towards the bottom, and as they approach the bottom, they curve back upwards. Here, the changes in speed of sound allow this layer of water to act as a wave guide. This layer of water is now called the SOFAR channel or Sound Frequency and Ranging channel, or deep sound channel or DSC. [7] Low-frequency sound can travel for much longer within the SOFAR channel without dissipating [8]

The discovery of the SOFAR channel has been influential in submarine warfare, as well as in biology. Whales are known to dive to the SOFAR channel where they can communicate with other whales several kilometers away.[9]

A similar behaviour of the speed of sound can be found in the atmosphere. The speed of sound generally decreases with altitude in the troposphere,

increases with altitude with the stratosphere, decreases with altitude in the mesosphere and finally increases in the thermosphere. [11] These fluctuations of the speed of sound in the atmosphere can similarly be mapped out using a linear relation between the speed of sound and altitude.

This same formulation can be applied to graded-index optics problems, if the speed of light as a function of a spatial variable changes in a linear fashion. For air, it might be useful to consider a system of air masses with alternating temperatures, which could be approximated by this linear model. An instance would be conditions in which mirages are formed. [12]

Chapter 2

Setting Up The Equations of Motion

2.1 An Analogy Between Rays and Particles

Analogies between rays and particles have been used to solve problems in ray propagation. [1] [13] [14] When a ray traverses a graded-index medium, a mechanical analogy could be supplied for its trajectory. It could act like a particle influenced by a potential energy function.

One approach is made by making an analogy between Fermat's principle and Lagrangian formulation. [1] For this paper, we will be using the relationship between a radial potential function and the refractive index calculated in Rangwala's paper[13].

Rangwala's paper starts with Fermat's principle. We assume a ray going from point $\vec{r}(\sigma_1)$ to $\vec{r}(\sigma_2)$. This traverses an optical path length $\int_{\sigma_1}^{\sigma_2} n(\vec{r}) d\sigma$. σ is some trajectory parameter that would correspond to time in mechanics. This path length is stationary with respect to very small variations of the path. Mathematically, this means that:

$$\delta \left\{ \int_{\sigma_1}^{\sigma_2} n(\vec{r}) \frac{ds}{d\sigma} d\sigma \right\} = 0 \quad (2.1)$$

Here, δ is any small variation over the paths, $n(\vec{r})$ is the refractive index and ds is:

$$ds = (dx_1^2 + dx_2^2 + dx_3^2)^{1/2} = |d\vec{r}| \quad (2.2)$$

where x_i , $i = 1, 2, 3$ are the position coordinates of the ray. These are a functions of σ .

(2.1) was compared with the Hamilton principle:

$$\delta \left\{ \int_{\sigma_1}^{\sigma_2} L(\vec{q}(\sigma), \dot{\vec{q}}(\sigma), \sigma) d\sigma \right\} = 0 \quad (2.3)$$

where q is a generalized coordinate.

We can now define an optical Lagrangian.

$$L_{opt} = n(\vec{r}) \frac{ds}{d\sigma} = n(\vec{r})v \quad (2.4)$$

From this Lagrangian, a linear momentum can be found:

$$p_i = \frac{\delta L}{\delta \dot{x}_i}, i = 1, 2, 3 \quad (2.5)$$

so:

$$\vec{p} = n(\vec{r}) \left(\frac{d\vec{r}}{d\sigma} \right) / \left(\frac{ds}{d\sigma} \right) = n(\vec{r}) \frac{d\vec{r}}{ds} \quad (2.6)$$

since $d\vec{r} = \hat{s}ds$, where $\hat{s}$ is a unit vector tangent to the ray trajectory at a certain point,

$$\vec{p} = n(\vec{r})\hat{s} \quad (2.7)$$

Now, With the Euler-Lagrange differential equation:

$$\frac{d}{d\sigma}\left(\frac{\delta L}{\delta \dot{x}_i}\right) - \frac{\delta L}{\delta x_i} = 0, i = 1, 2, 3 \quad (2.8)$$

We use equation (2.6) after moving the second term in (2.7) to the right hand side:

$$\frac{d\vec{p}}{d\sigma} = \nabla L \quad (2.9)$$

Since this is the form of a force equation, we can define a potential function such that:

$$V(\vec{r}, v) = -L(\vec{r}, v) \quad (2.10)$$

Next, we stipulate that:

$$\sigma = \int \frac{ds}{n} \quad (2.11)$$

such that the refractive index:

$$n = \frac{ds}{d\sigma} \quad (2.12)$$

We can now solve for ∇L with the identity in (2.12), which gives us:

$$\nabla = \nabla(n(\vec{r})v) = \nabla(n(\vec{r}))\frac{ds}{d\sigma} = \nabla(n(\vec{r}))n(\vec{r}) = \frac{1}{2}\nabla(n^2(\vec{r})) \quad (2.13)$$

Thus, we can write equation (2.9) and its resulting potential as:

$$\frac{d\vec{p}}{d\sigma} = \nabla\left(-\frac{1}{2}n^2(\vec{r})\right) \quad (2.14)$$

$$V(\vec{r}) = -\frac{1}{2}n^2(\vec{r}) + \text{const.} \quad (2.15)$$

In this analogy, we will scale the value of mass for as 1.

2.2 For The RHO Profile

In our current system, we have a region $a < r < r_0$ where the speed of sound decreases and the index of refraction increases as r increases. We also have a region $r_0 < r < b$ where the speed of sound increases and the index of refraction increases as r increases.

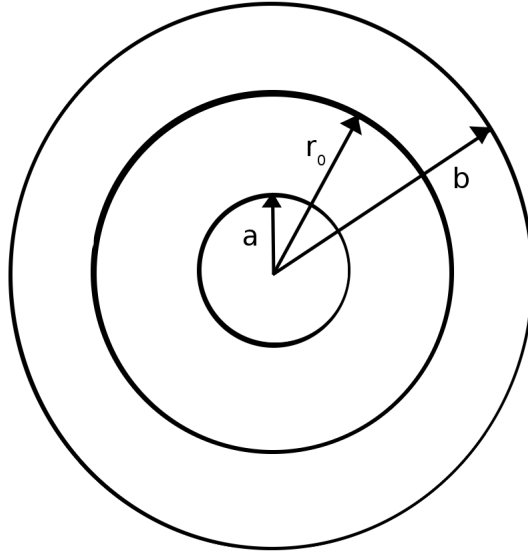


Figure 2.1: The radial refractive index

In the region $r_0 < r < b$, we use our $n^2(r)$ profile and we recall the analogy:

$$-\frac{1}{2}n^2(r) + const. = \frac{1}{2}(-n_b - C_1 + C_2r^2) + const. = V(r) \quad (2.16)$$

Note that n_b is the index of refraction beyond the outer boundary of the medium. Notice that the resulting potential looks like a radial harmonic oscillator potential, which would generate closed orbits. We can now differentiate the potential and we have a central force:

$$F(r) = -\frac{d}{dt}V(r) = -C_2r$$

If we assume $C_2 > 0$, this gives the following equations of motion:

$$x(t) = x_{entry} \cos \omega_1 t + \frac{\dot{x}_{entry}}{\omega_1} \sin \omega_1 t \quad (2.17)$$

$$y(t) = y_{entry} \cos \omega_1 t + \frac{\dot{y}_{entry}}{\omega_1} \sin \omega_1 t \quad (2.18)$$

where ω_1 is $\sqrt{C_2}$

Note that the final trajectory is independent of either n_b or C_1

In the region $a < r < r_0$, we can also get a solvable radial potential:

$$-\frac{1}{2}n^2(r) + const. = -\frac{1}{2}(n_a + C_3r^2) + const. = V(r) \quad (2.19)$$

Note that n_a is the index of refraction beyond the inner boundary of the medium. This is an inverse radial harmonic potential. Differentiating the potential, we have a central force:

$$F(r) = -\frac{d}{dt}V(r) = C_3r$$

Which will have the following equations of motion:

$$x(t) = x_{entry} \sinh \omega_2 t + \frac{\dot{x}_{entry}}{\omega_2} \cosh \omega_2 t \quad (2.20)$$

$$y(t) = y_{entry} \sinh \omega_2 t + \frac{\dot{y}_{entry}}{\omega_2} \cosh \omega_2 t \quad (2.21)$$

where ω_2 is $\sqrt{C_3}$

Note that the effects of this graded index are truncated at the region $r > \sqrt{\frac{C_1}{C_2}}$, because beyond this region, the preceding potential, $V(r) = C_1 - C_2 r^2$ would start becoming negative, and thus produce imaginary refractive indices.

Chapter 3

Results and Plots

These are the equations of motion for the RHO profile which we have solved in Chapter 2, which result in oscillatory motion of the trajectory about the boundary:

For $r_0 < r < b$:

$$x(t) = x_{entry} \cos \omega_1 t + \frac{\dot{x}_{entry}}{\omega_1} \sin \omega_1 t \quad (3.1)$$

$$y(t) = y_{entry} \cos \omega_1 t + \frac{\dot{y}_{entry}}{\omega_1} \sin \omega_1 t \quad (3.2)$$

where ω_1 is $\sqrt{C_2}$

For $a < r < r_0$:

$$x(t) = x_{entry} \sinh \omega_2 t + \frac{\dot{x}_{entry}}{\omega_2} \cosh \omega_2 t \quad (3.3)$$

$$y(t) = y_{entry} \sinh \omega_2 t + \frac{\dot{y}_{entry}}{\omega_2} \cosh \omega_2 t \quad (3.4)$$

where ω_2 is $\sqrt{C_3}$

The arbitrary parameters for the inner boundary and outer boundary were chosen: $a = 0.2r_0$ and $b = 2r_0$ scaled in terms of r_0 .

We will define the launch angle in Figure 3.1, the angle that the initial velocity of the ray makes with a vector perpendicular to the radial vector.

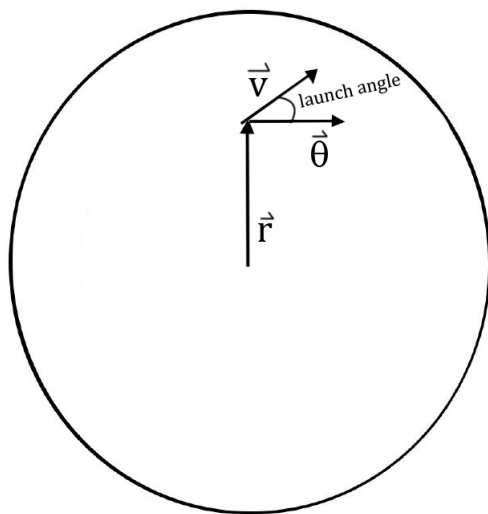


Figure 3.1: An illustration of the launch angle

3.1 Results for Varying r_i and launch angle

Figures 3.2 and 3.3 will show the ray trajectories for varying initial launch position or r_i . The launch angles for these figures were set to $\theta_i = 0$. Figures 3.4 and 3.5 will show ray trajectories for varying launch angles, with the initial position set to $1.2r_0$. Figures 3.6 to 3.11 show how the initial position and the launch angle vary the amplitude, wavelength and tangential speed of the ray trajectory. All of these values are scaled to r_0 . The following plots also set $\omega_1 = \omega_2 = 1$.

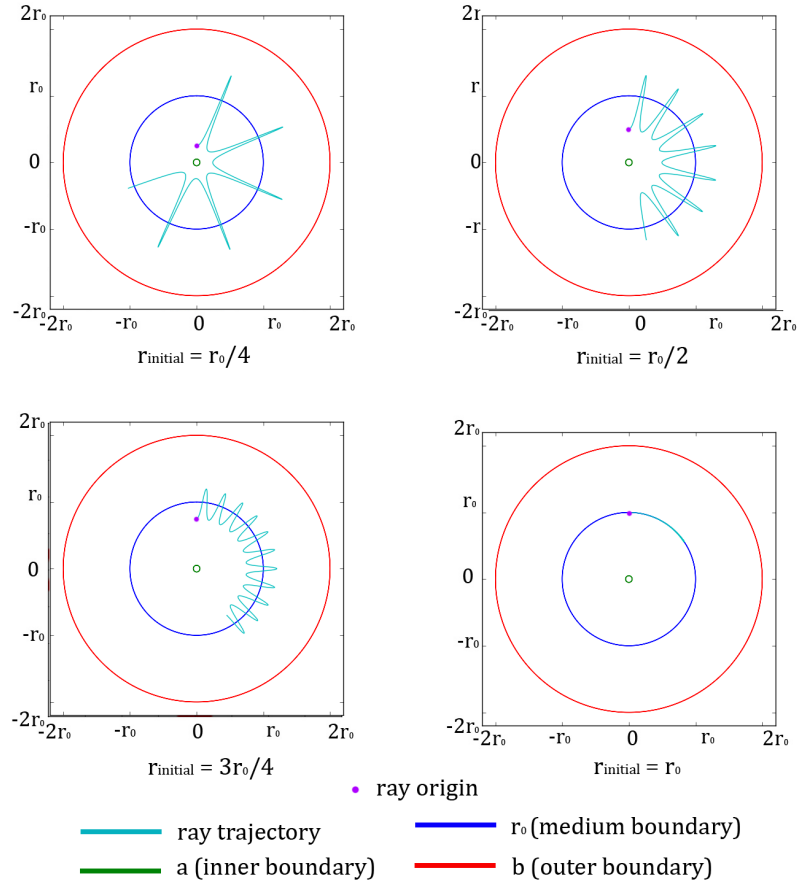


Figure 3.2: Ray trajectories for varying r_i

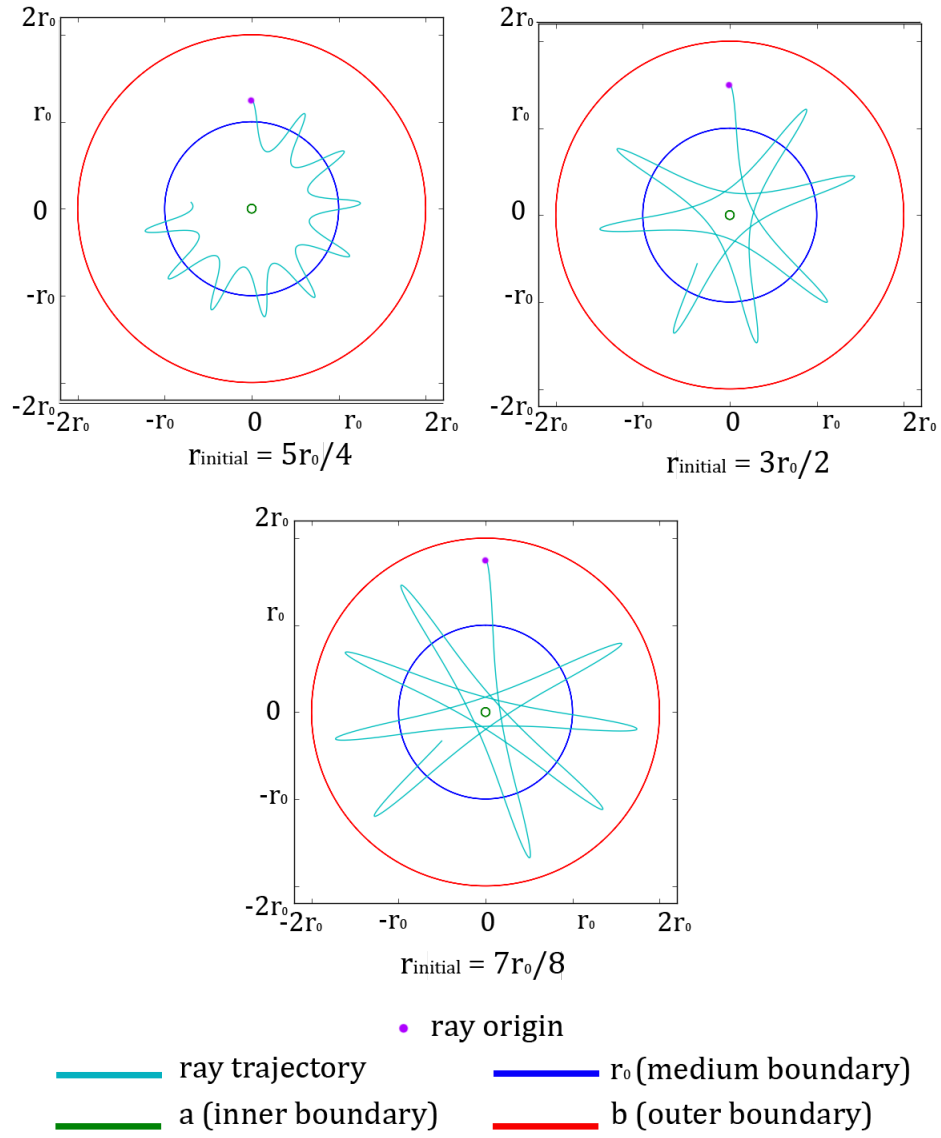


Figure 3.3: Ray trajectories for varying r_i

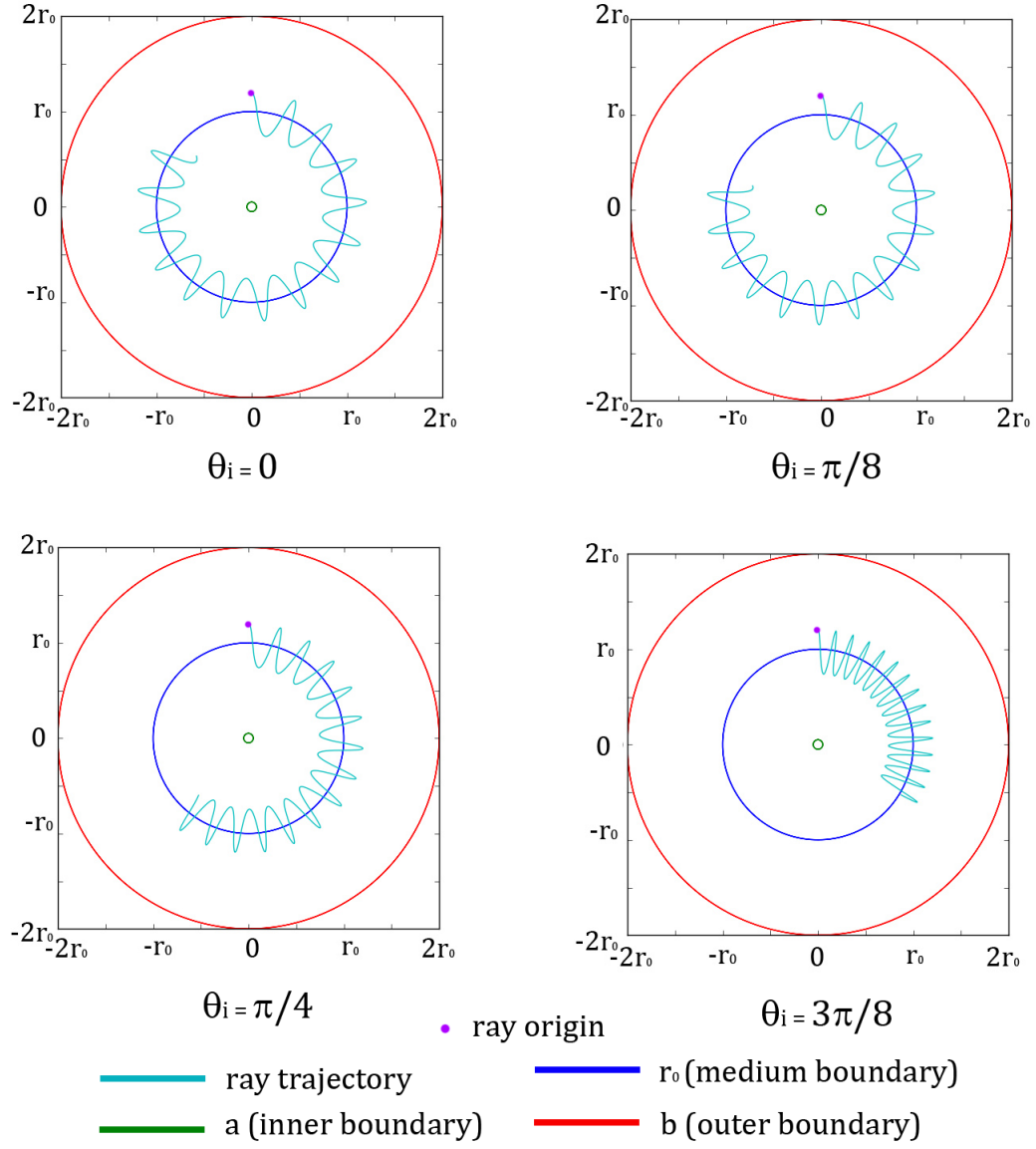


Figure 3.4: Ray trajectories for varying launch angles

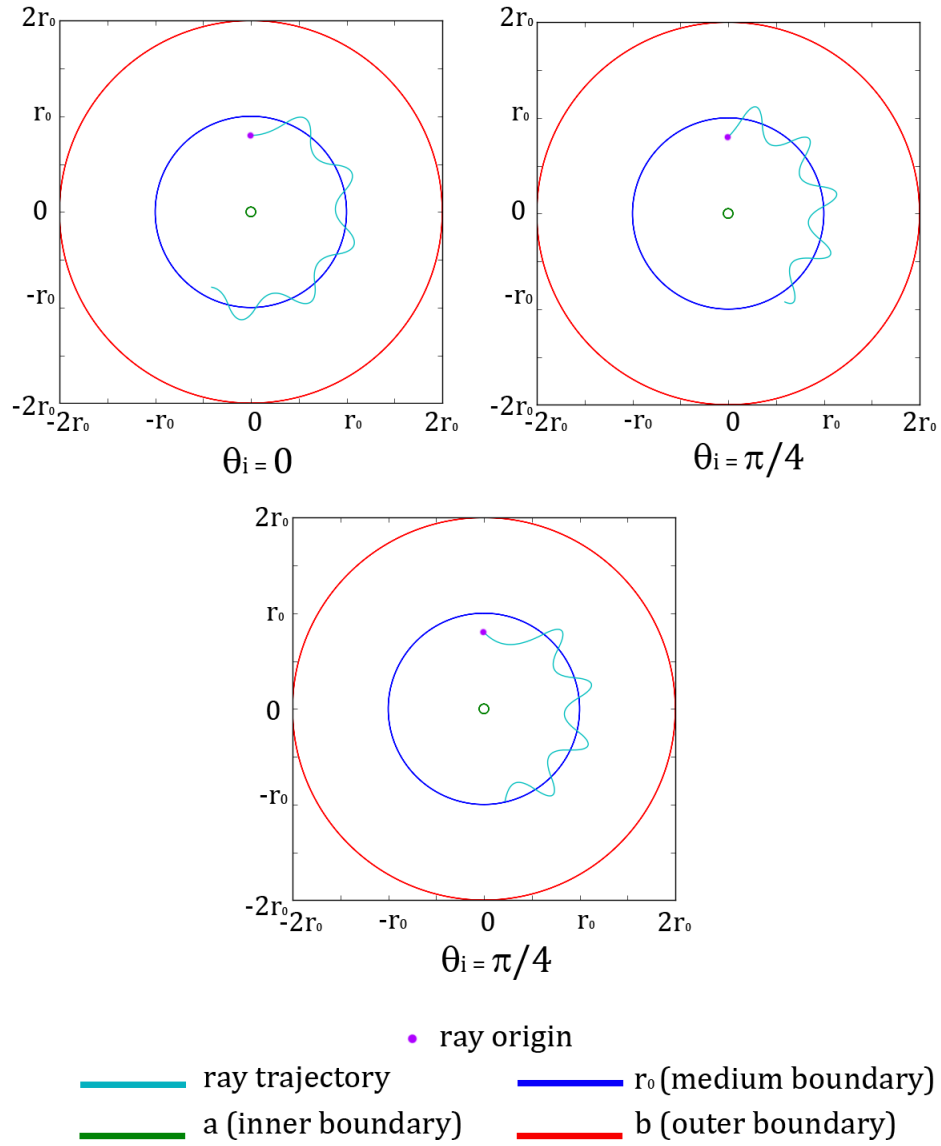


Figure 3.5: Ray trajectories for varying launch angles

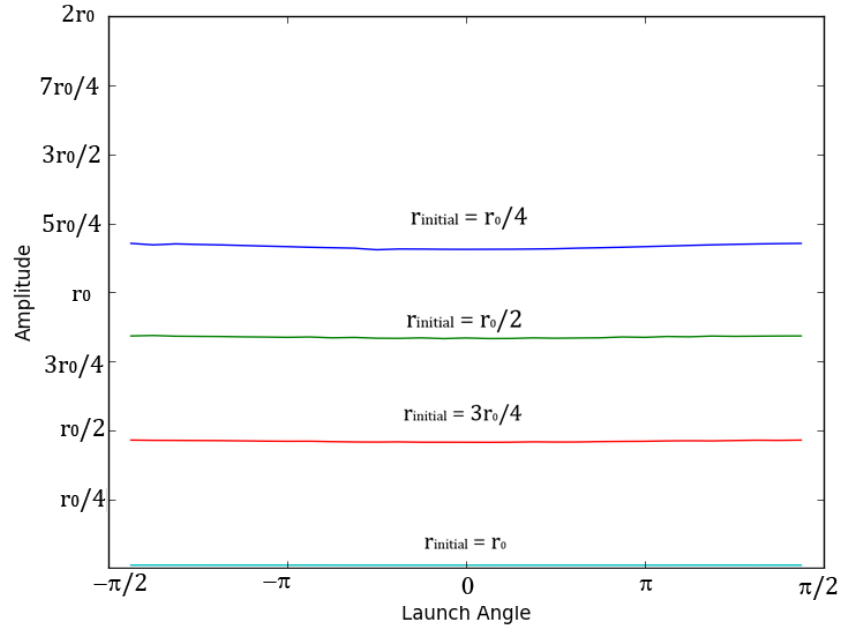


Figure 3.6: Plot showing dependence of amplitude of ray trajectory on launch angle and r_i for the region $r_{\text{initial}} < r_0$

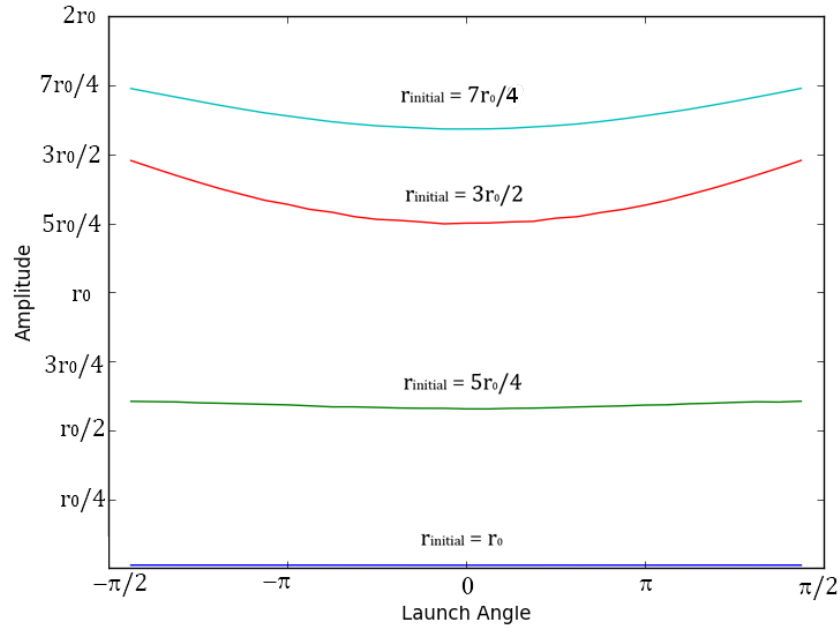


Figure 3.7: Plot showing dependence of amplitude of ray trajectory on launch angle and r_i for the region $r_{\text{initial}} > r_0$

In Figures 3.6 to 3.7, the amplitude was determined by gathering the minimum and maximum distances from the center of the medium that the ray traverses and then taking their difference.

The amplitude of the ray trajectory is dependent mostly on the initial position of the ray. The farther the initial position is for r_0 , the greater the amplitude. This is consistent with the properties of a harmonic oscillator. The farther the initial position is from the axis of oscillation, the greater the initial potential energy and the resulting amplitude is greater.

The θ_i affects the amplitude minimally. However, The greatest amplitudes occur when the launch angles approach $-\pi/2$ and $\pi/2$. The amplitude reaches a minimum at when the launch angle is 0. When the launch angles are close to $-\pi/2$ and $\pi/2$, this means that the initial velocity of the ray is close to becoming parallel with the radial vector. This means that the initial velocity contributes kinetic energy to the total energy of the ray, thus increasing the amplitude. When the launch angle is 0, this means that the initial velocity only contributes only to the initial angular speed of the ray, and nothing along the radial axis. It does not affect the amplitude this way.

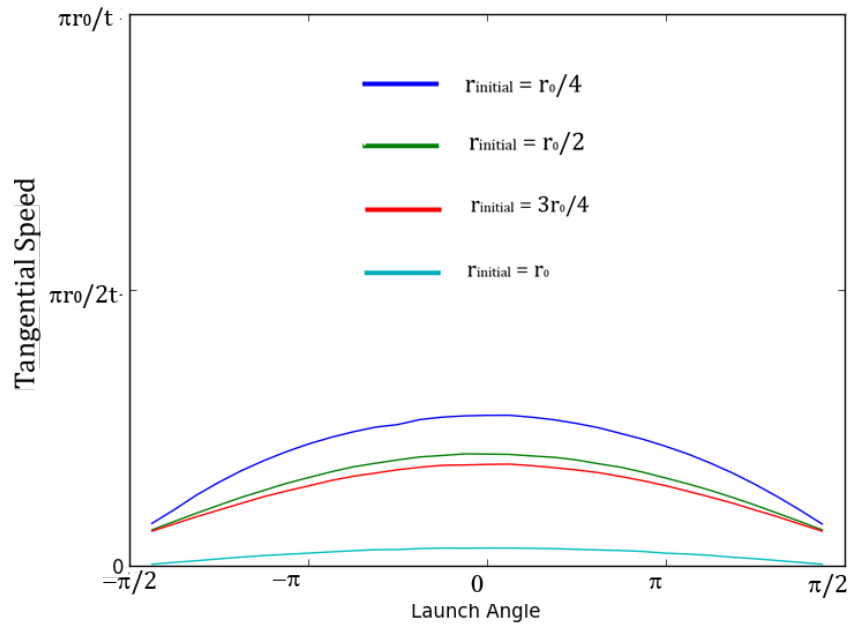


Figure 3.8: Plot showing dependence of tangential speed of ray trajectory on launch angle and r_i for the region $r_{\text{initial}} < r_0$

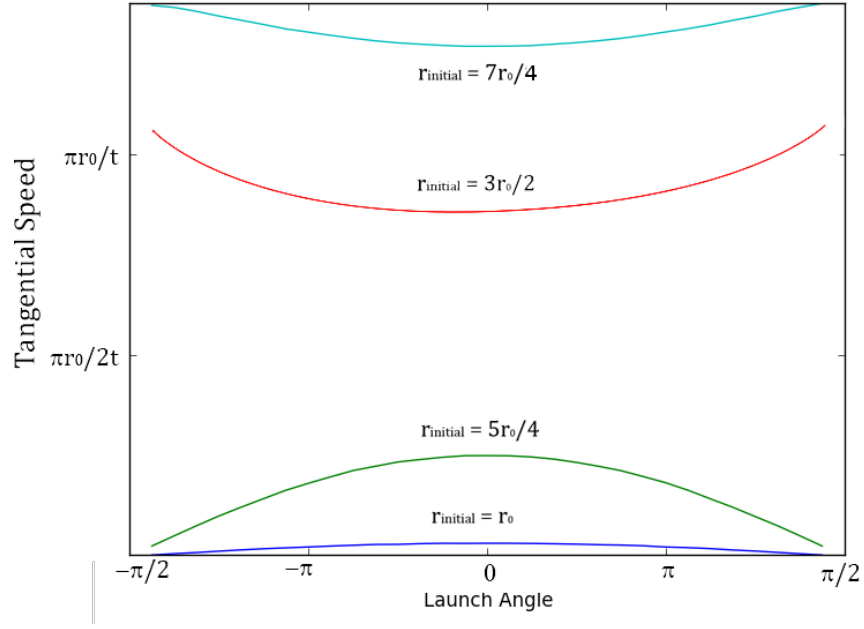


Figure 3.9: Plot showing dependence of tangential speed of ray trajectory on launch angle and r_i for the region $r_{\text{initial}} > r_0$

In figures 3.8 and 3.9, the tangential speed is the ratio of $r_0\theta$ to time was taken. θ is the angular distance traversed by the ray through the medium. The tangential speed was determined by propagating the rays at constant time, and taking the final $r_0\theta$ traversed by the ray.

We can see that the tangential speed generally decreases as the initial position is nearer to r_0 . This makes sense because since we are trying to plot a refractive index profile that is increasing up to r_0 then decreasing beyond r_0 , we can expect that the ray speed decreases and reaches a minimum as the ray approaches r_0 .

For the launch angles, we can see that the maximum tangential speed occurs when the θ_i is 0. This is an expected result because the initial velocity contributes most to the angular speed of the ray at this angle. However, as the distance from the center increases, a shift in how the launch angle influences the tangential velocity occurs. As the distance from the center increases, the ray gradually takes on different oscillatory behaviour. It no longer exhibits wave-like properties and it is able to cross a large distance of the inner medium. We can see this at Figures 3.2 to 3.3. This transition occurs when the exit angle of the ray is greater than $\pi/2$. We will define the exit angle, θ_{exit} , of the ray as the angular distance between the velocity vector of the ray when it enters the inner medium, and the velocity vector when it exits the inner medium, in a single oscillation. Beyond this point, the ability of the ray to propagate along the radial axis increases its range and its propagation along the angular axis decreases its range.

The launch angles close to $-\pi/2$ and $\pi/2$ now result in maximum tangential speed, while the launch angle at 0 results in the minimum tangential speed.

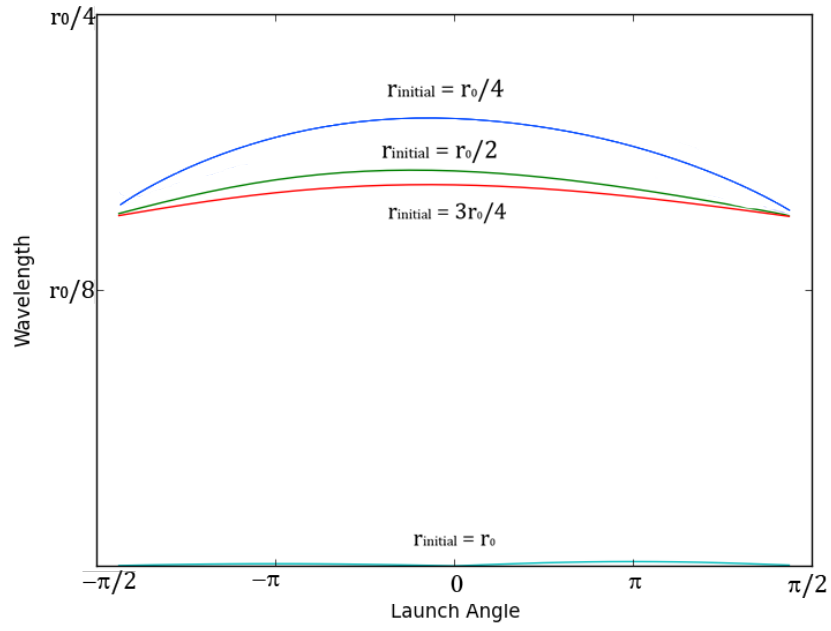


Figure 3.10: Plot showing dependence of wavelength of ray trajectory on launch angle and r_i for the region $r_{\text{initial}} < r_0$

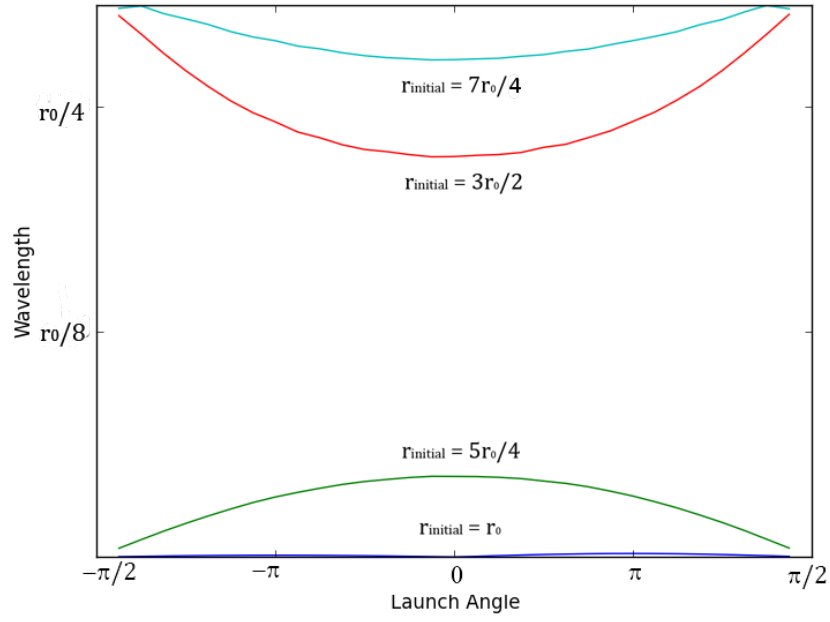


Figure 3.11: Plot showing dependence of wavelength of ray trajectory on launch angle and r_i for the region $r_{\text{initial}} > r_0$

In figures 3.10 to 3.11, we see plots of the wavelength. This was determined by taking the $r\theta$ traversed by the ray between the first and third instances when it crosses the boundary. We can see that the changes in the wavelength due to θ_i and r_i correspond to the changes in initial velocity. Both are minimum when the r_i is close to r_0 . Both are maximum when θ_i is 0, until the ray's exit angle is greater than $\pi/2$, in which case the maxima occur at $\pm\pi/2$.

These changes for wavelength occur for the same reasons as the changes in tangential speed occur. Wavelength is minimum at $r_i = r_0$ because since ray speed is slowest in this region and amplitude is also minimum in this region, we would expect that the wavelength should also be minimum in this region. Maxima occur for $\theta_i = 0$ because the initial velocity contributes most to the angular speed of the ray. When the $\theta_{exit} > \pi/2$, The radial speed now becomes the determining factor in wavelength and tangential speed, thus the maxima occur near $\theta_{exit} = \pm\pi/2$.

3.2 Results for Varying ω values

In the following graphs we will vary the omega values. We can see ray trajectories plotted with various ω in Figure 3.12, and how the ω values affect the amplitude, velocity, and tangential speed of the ray trajectory in Figures 3.13-3.15. These ray trajectories have the initial conditions $r_{initial} = 1.2r_0$ with launch angles of 0. A set of figures 3.16-3.19 show how changes in the ω values affect the trajectory when the initial position is beneath the boundary. We set $r_i = 0.8r_0$ for these plots. We will refer to ω_1 and ω_2 as ω_{outer} and ω_{inner} .

It is also important to reiterate that the $\omega > 0$.

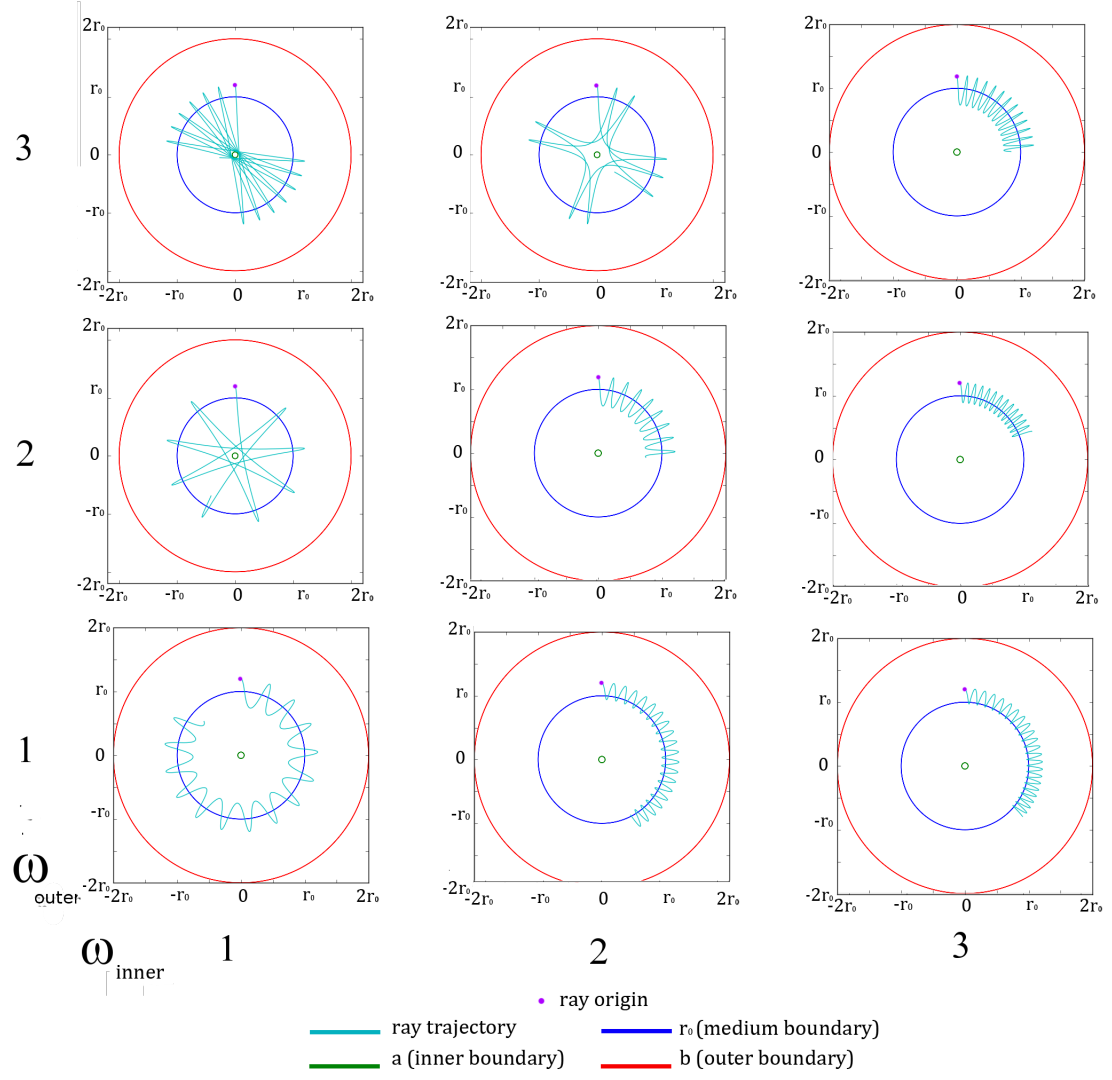


Figure 3.12: Ray trajectories resulting from varying ω values for $r_i = 1.2r_0$.

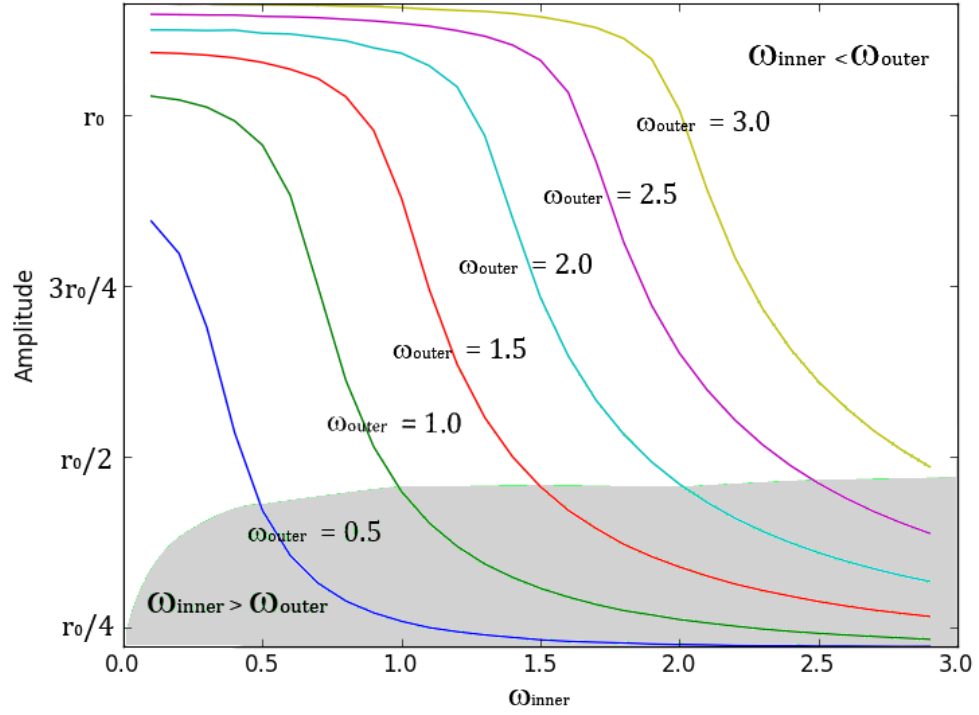


Figure 3.13: Plots showing the dependence of amplitude on the ω values for $r_i = 1.2r_0$. The gray region represents the region where $\omega_{inner} > \omega_{outer}$ while the white region represents $\omega_{inner} < \omega_{outer}$

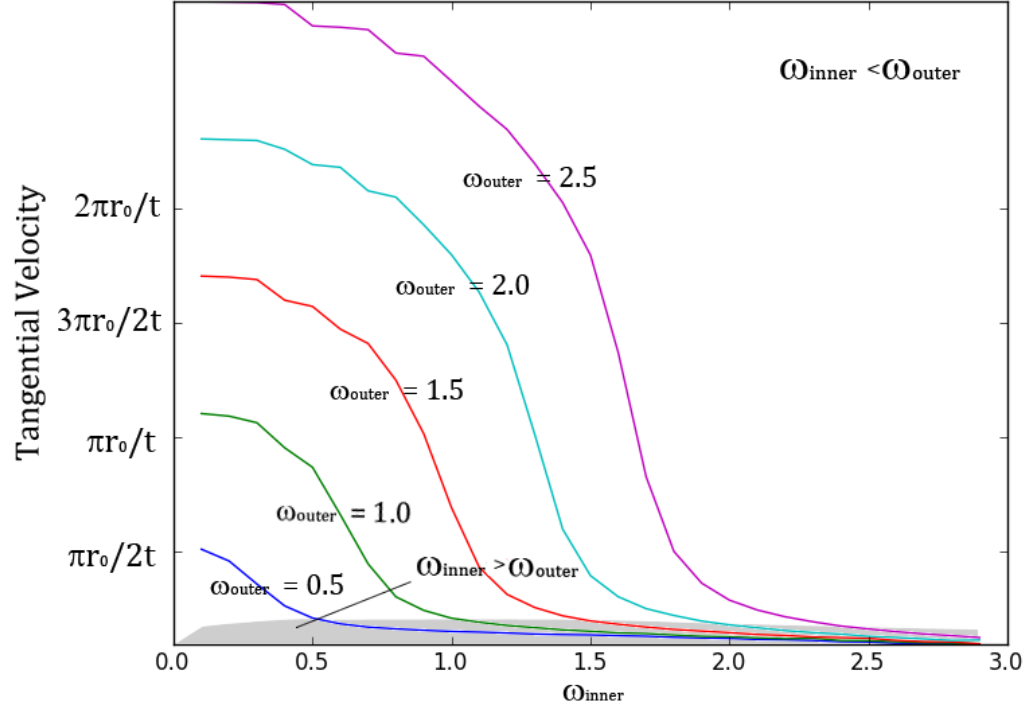


Figure 3.14: Plots showing the dependence of tangential speed on the ω values for $r_i = 1.2r_0$. The gray region represents the region where $\omega_{inner} > \omega_{outer}$ while the white region represents $\omega_{inner} < \omega_{outer}$

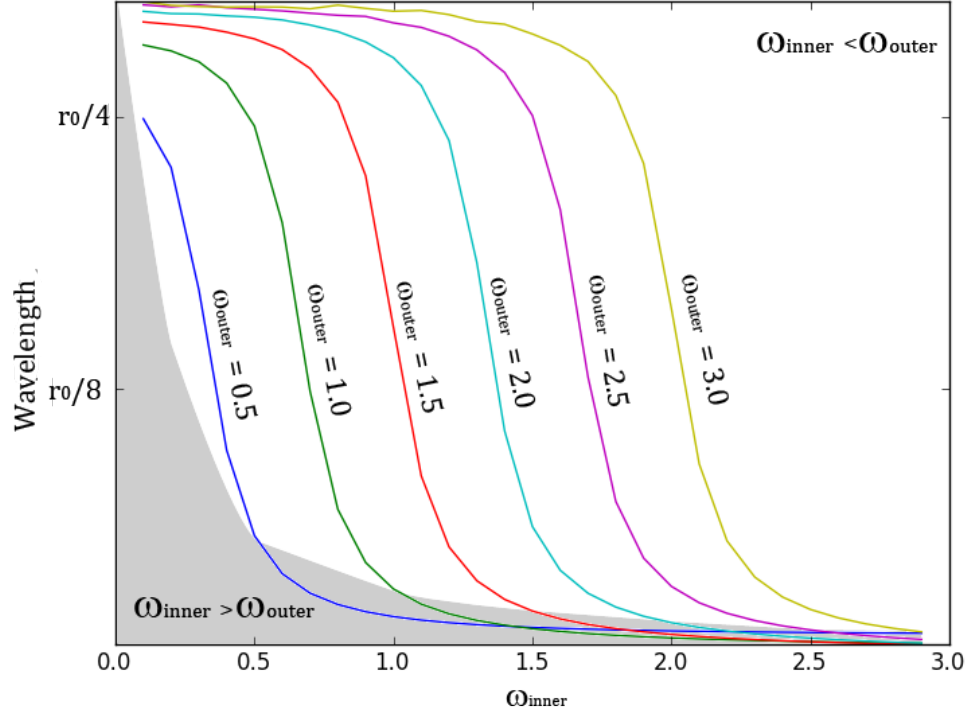


Figure 3.15: Plots showing the dependence of wavelength on the ω values for $r_i = 1.2r_0$. The gray region represents the region where $\omega_{inner} > \omega_{outer}$ while the white region represents $\omega_{inner} < \omega_{outer}$

The ω_{outer} corresponds to $\sqrt{C_2}$ in our equations, which the coefficient of the force equation. Higher ω_{outer} means that the attractive force towards r_0 is also higher.

The ω_{inner} corresponds to $\sqrt{C_3}$ in our equations, a coefficient of the force equation for the inner region of the medium. A higher ω_{inner} corresponds to a greater repulsive force when the ray is in the inner region.

When the ray is launched from above the boundary, for the case of $\omega_{inner} < \omega_{outer}$, the ray tends to follow star-like trajectories, as its θ_{exit} is now greater than $\pi/2$. Since increasing the ω_{outer} would in effect increase its initial energy, It would result in larger amplitudes. Because the ray follows these trajectories in this medium, a greater ω_{outer} also increases the tangential velocity and wavelength of the ray for this case.

However, as the ω_{inner} matches the value of ω_{outer} , higher values of ω_{outer} produce shorter wavelengths instead of longer wavelengths, as we can see in figure 3.15. This occurs because when the forces in both media are stronger, more rapid oscillation occurs about the boundary, and shorter wavelengths are produced.

In the case when $\omega_{inner} > \omega_{outer}$ the repulsive force in the inner medium is greater. The greater the ω_{inner} , the less time the ray makes shorter and small oscillations in the inner medium, and for higher values of $\omega_{inner}/\omega_{outer}$, the ray produces a skipping motion where it barely enters the inner medium.

Greater values of ω_{inner} produce lower values of amplitude, tangential speed, and wavelength. We can notice that larger values of ω_{inner} result in shorter wavelengths and smaller amplitudes within the inner medium. The amplitude and wavelength of the ray trajectory decrease due to the smaller oscillations. Larger values of ω_{inner} mean that the velocity of the ray decreases more quickly as it penetrates the inner medium. By the time

it exits the inner region, it would have traversed a small angular distance, which would correspond to a smaller tangential speed.

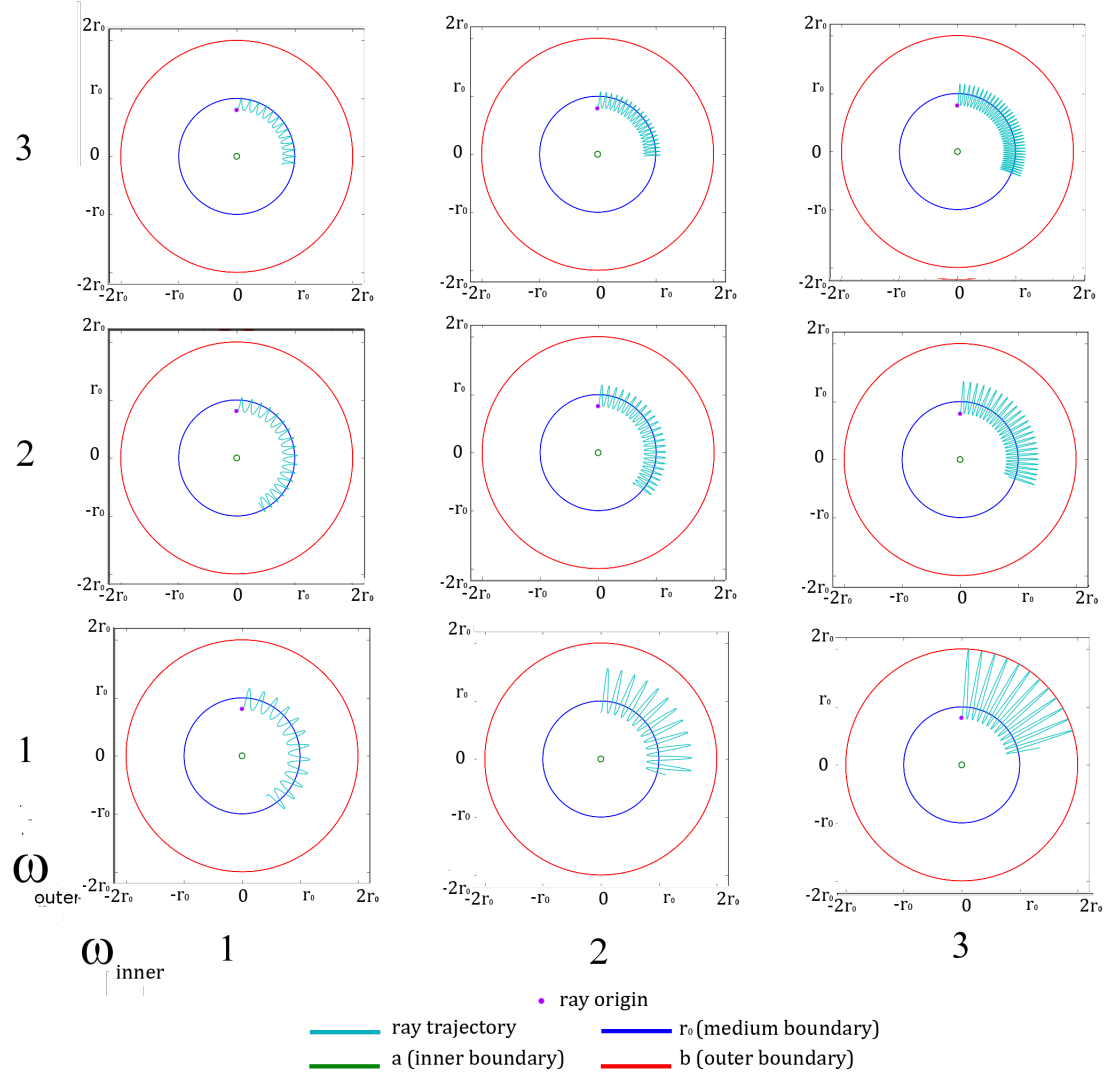


Figure 3.16: Ray trajectories resulting from varying ω values for $r_i = 0.8r_0$.

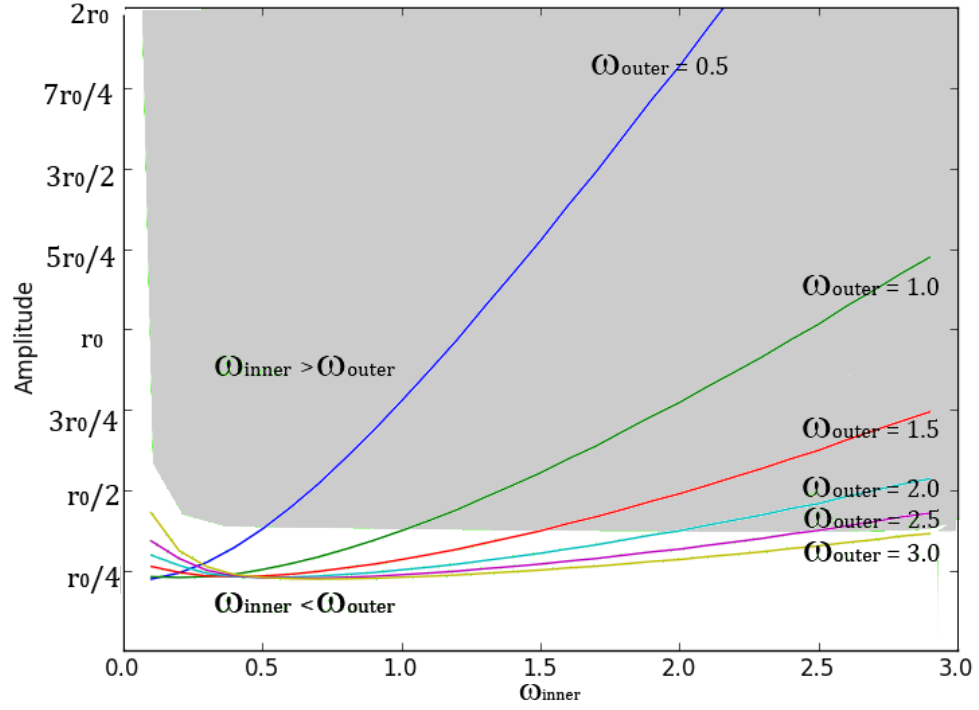


Figure 3.17: Plots showing the dependence of amplitude on the ω values for $r_i = 0.8r_0$. The gray region represents the region where $\omega_{inner} > \omega_{outer}$ while the white region represents $\omega_{inner} < \omega_{outer}$

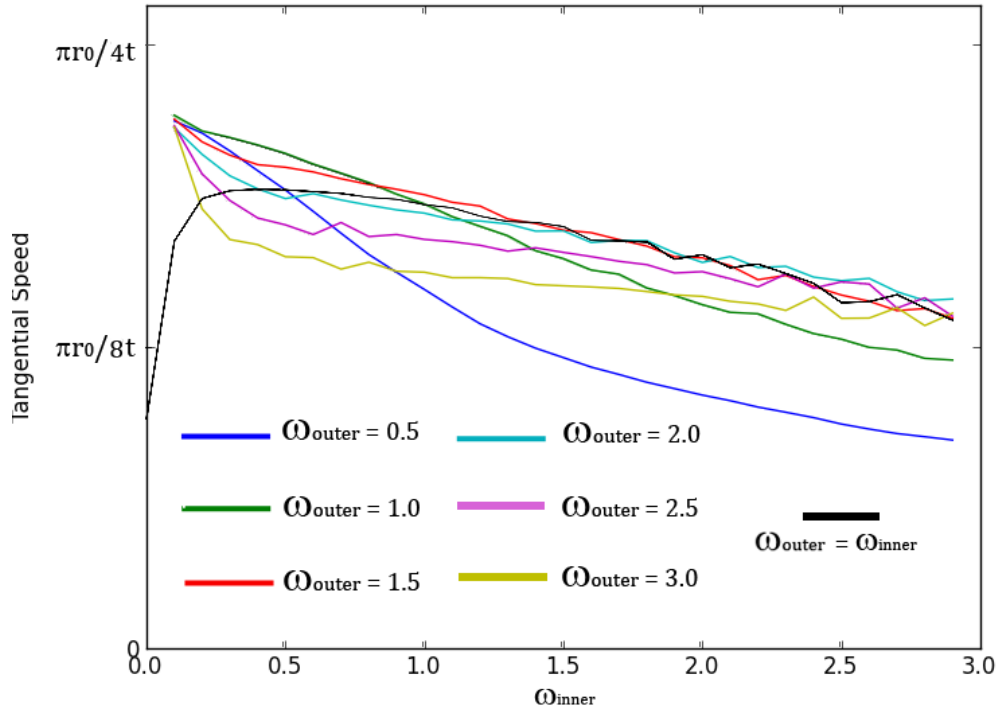


Figure 3.18: Plots showing the dependence of tangential speed on the ω values for $r_i = 0.8r_0$. The black line represents where $\omega_{inner} = \omega_{outer}$.

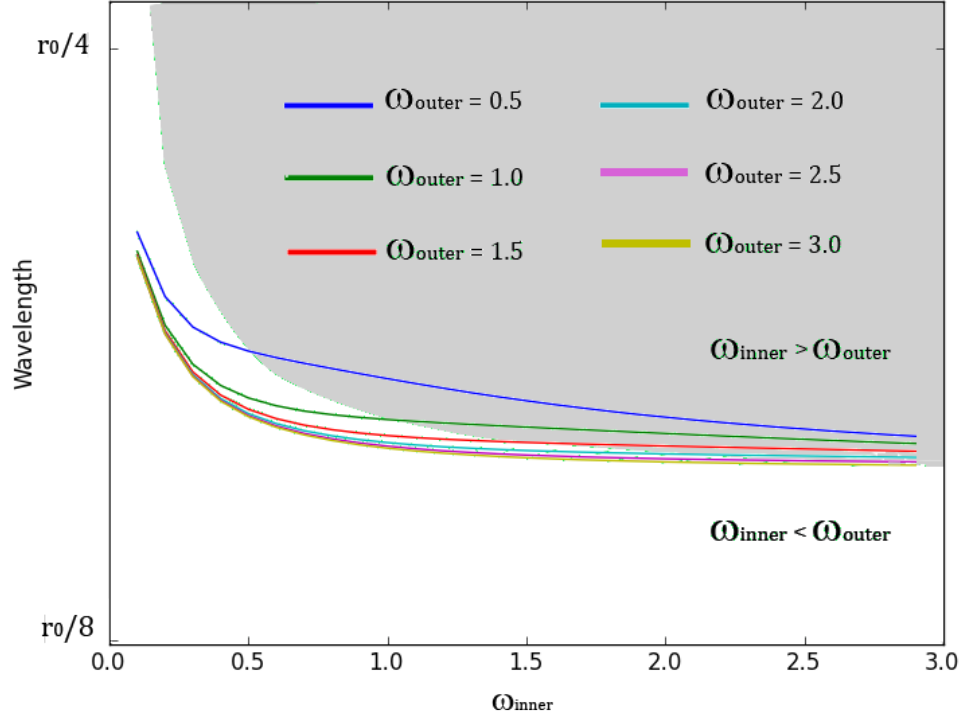


Figure 3.19: Plots showing the dependence of wavelength on the ω values for $r_i = 0.8r_0$. The gray region represents the region where $\omega_{inner} > \omega_{outer}$ while the white region represents $\omega_{inner} < \omega_{outer}$

When the ray is launched from below the boundary, for the case of $\omega_{outer} < \omega_{inner}$, the ray produces large oscillations in the outer medium. An increase in the ω_{inner} means that the ray is propelled more greatly by the inverse radial harmonic oscillator or IRHO force. A decrease in ω_{outer} corresponds to less resistance by the radial harmonic oscillator or RHO force. These correspond to greater amplitude for this case, as shown by Figure 3.17. We notice in Figure 3.18, this case of $\omega_{outer} < \omega_{inner}$ produces smaller tangential velocity of the ray trajectory as ω_{outer} is decreased. This is because smaller values of ω_{outer} make the ray travel larger distances along the radial axis. However, it travels a short angular distance by the time it completes a cycle in the outer boundary.

As the ω_{outer} approaches the value of ω_{inner} , higher values of ω_{outer} , similar oscillatory behaviour occurs for when the ray is launched outside the boundary. There are more rapid oscillations about r_0 , resulting in shorter wavelengths. For these reasons, we can see that in figure 3.19, the wavelength decreases as ω_{outer} decreases and as ω_{inner} increases.

Compared to the previous case when $\omega_{outer} < \omega_{inner}$, as the two ω values match, the ray would have to traverse less distance in the outer medium for this case. It would be able to cover more angular distance, and thus the angular velocity increases as the values match.

For the case $\omega_{outer} > \omega_{inner}$, a similar skipping motion occurs within the inner region of the medium. In this region, the tangential velocity of the medium is slower with greater ω_{outer} , similar to the behaviour of the skipping motion in the outer medium.

Also note that when ω_{inner} is very small in this region, increasing the ω_{outer} increases the amplitude of the ray trajectory. A small ω_{inner} means that there

is a smaller force influencing the rays movement in the inner medium, and a greater ω_{outer} makes the ray trajectory trace larger arcs inside the inner medium.

Chapter 4

Conclusion and Recommendation

In this thesis we have shown that using an analogy between rays and particles would determine the trajectories of a spherically symmetric refractive index profile:

$$n^2(r) = \begin{cases} n_a + C_1 - C_2 r^2 & , b > r > r_0 \\ n_b + C_3 r^2 & , a < r \leq r_0 \\ n_b & , r > \sqrt{\frac{C_1}{C_2}} \end{cases} \quad (4.1)$$

where $C_1 = C_3 r_0^2 + n_b - n_a$

which we have found analytically. This refractive index increases and reaches a maximum at r_0 and then decreases. Thus, this profile acts as a spherical waveguide, producing ray trajectories that would oscillate around r_0 .

We have shown how varying the initial parameters for the RHO profile affect its trajectories.

The pieces of these refractive indices can be rearranged and manipulated to describe other systems. The two pieces can be switched, if the refractive index reaches a minimum at r_0 instead of a maximum. They can also be stacked, where instead of the ray exiting out the surface, it would enter

another piecewise system. This would be applicable in environments like the earth's atmosphere, where the temperature fluctuates frequently as different layers are entered.

It might also be useful to see how wind and water currents would affect the trajectories, which are prevalent in oceanic and atmospheric SOFAR channels. The rotation of the medium can also be considered.

Appendix A

Ray Pulse Evolution for the RHO profile

We now also show some preliminary results, where in a ray pulse is emitted at a certain $r_{initial}$ and then propagated through ωt increments so we can see how a pulse might evolve over time.

The pulses were constructed by propagating rays with different launch angles from the $r_{initial}$. Their final positions after propagating for a given value of ωt were collected and plotted. In effect, this pulse is the wavefront of an ray emission at a certain $r_{initial}$. In these plots, the magenta lines correspond to the pulse at a given time.

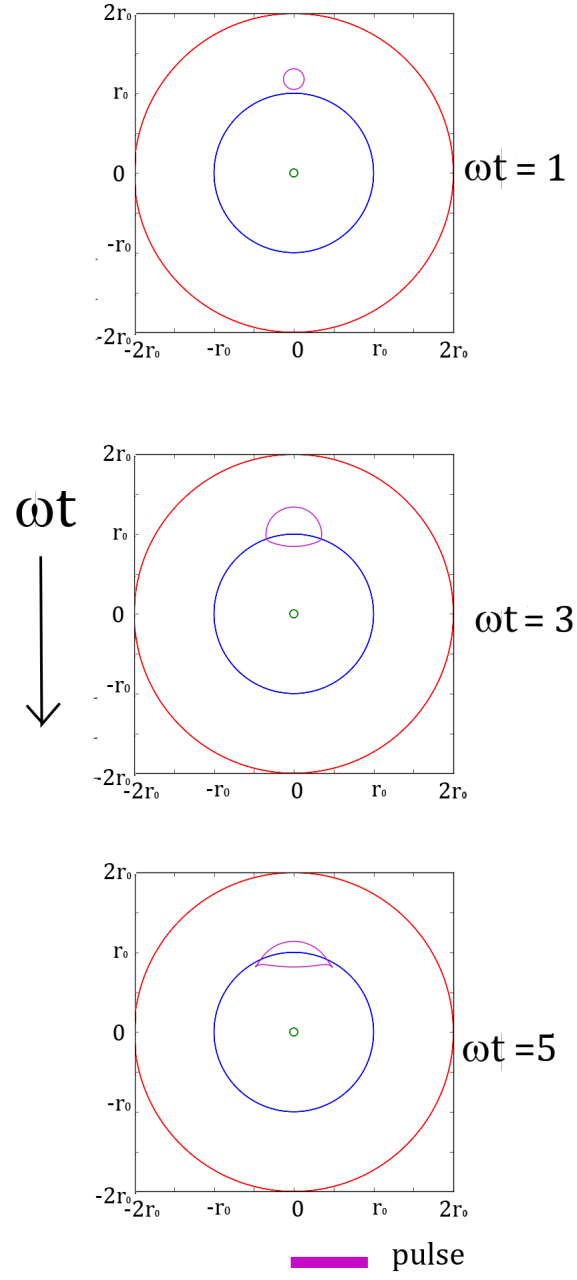


Figure A.1: Evolution for pulse emitted at $r_{initial} = 1.2r_0$.

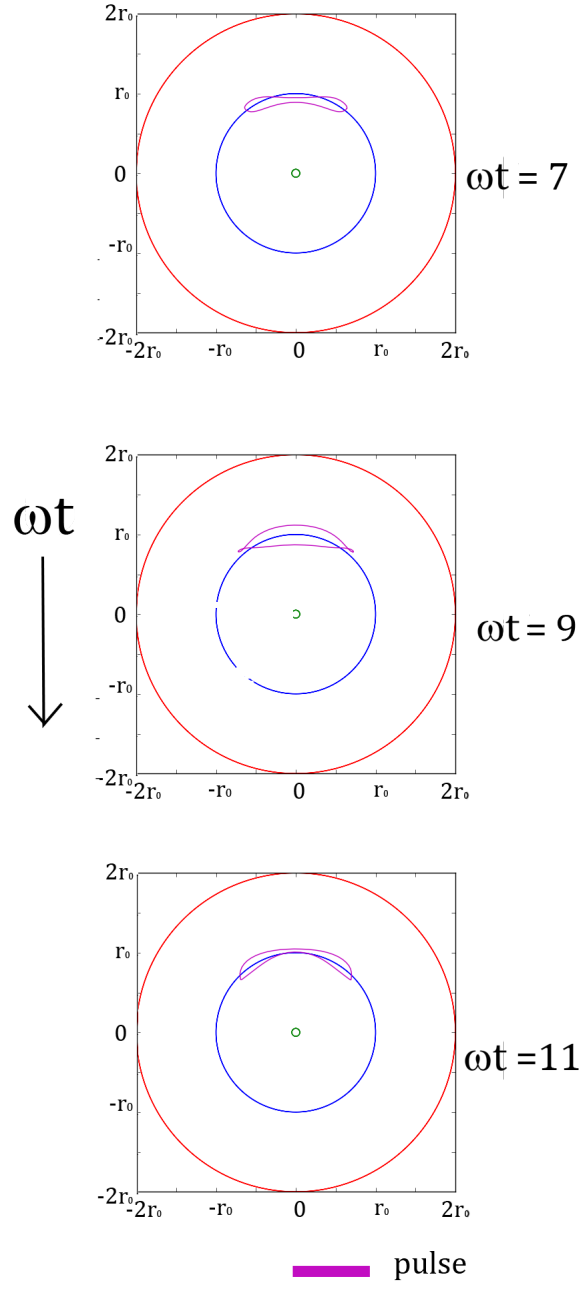


Figure A.2: Evolution for pulse emitted at $r_{initial} = 1.2r_0$.

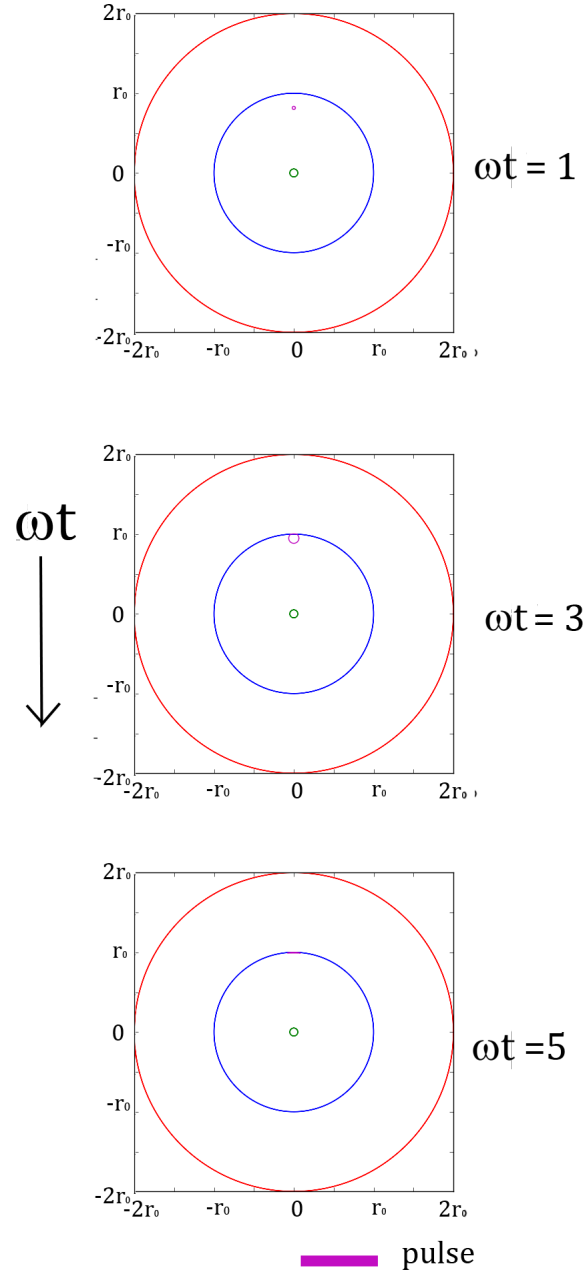


Figure A.3: Evolution for pulse emitted at $r_{initial} = 0.8r_0$.

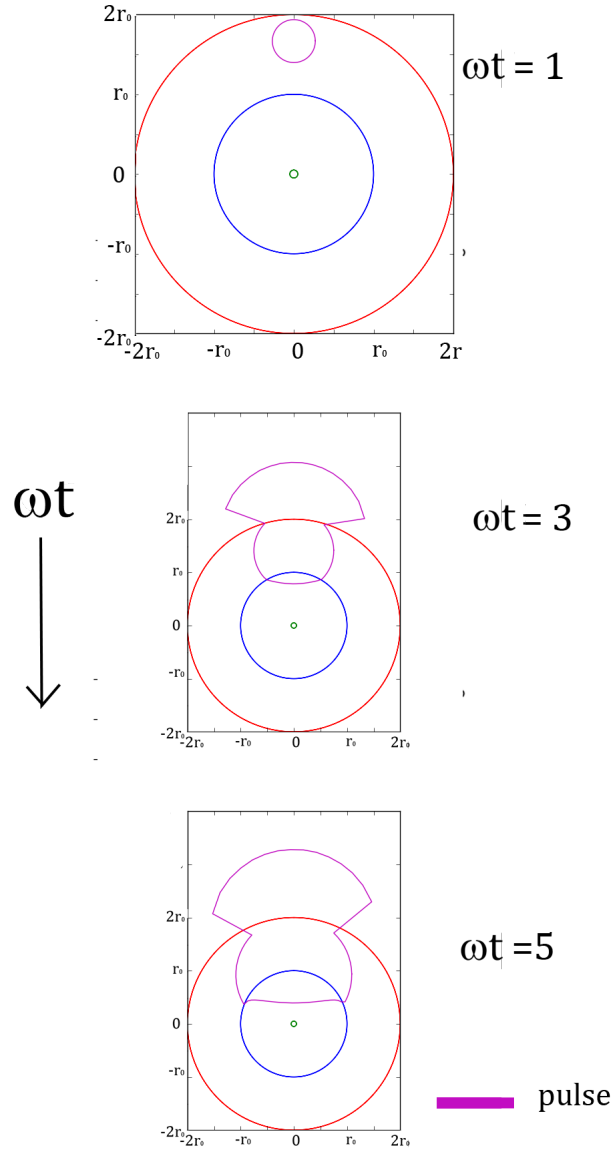


Figure A.4: Evolution for pulse emitted at $r_{initial} = 1.7r_0$.

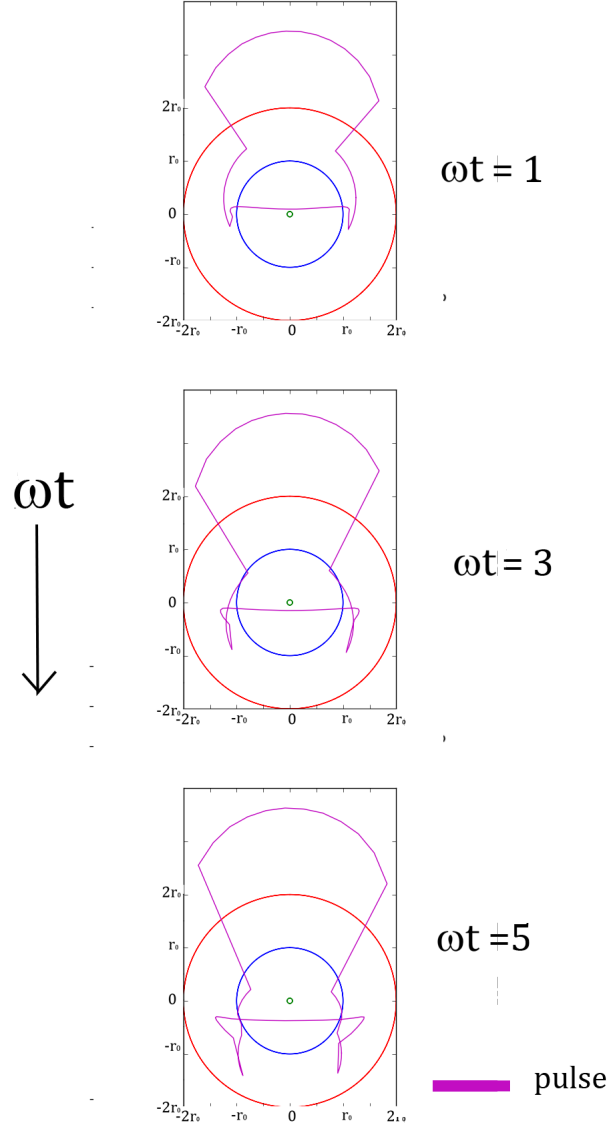


Figure A.5: Evolution for pulse emitted at $r_{initial} = 1.7r_0$.

In these plots we can see that pulses emitted near the boundary are absorbed by it. Pulses emitted beneath the boundary, $r_{initial} < r_0$, rise to the boundary and then remain there. Pulses emitted near and above the boundary are also absorbed by the boundary. However, when the pulse emission is close to the surface, more rays are able to escape the medium.

It is also important to note that all rays with a trajectory perpendicular to the radial vector will not be influenced by the medium, and these rays will always travel a straight path away or to the center of the medium.

Appendix B

The CVG Profile

B.1 Definition

We will consider a piecewise ray speed profile given by:

$$c(r) = \begin{cases} c_0 + m_1(r - r_0) & , b > r > r_0 \\ c_0 + m_2(r_0 - r) & , a < r \leq r_0 \end{cases} \quad (\text{B.1})$$

Here, c_0 is the minimum speed of the ray in the system. r_0 is the distance at which the speed of the ray is minimum in the medium. a and b indicate the boundaries of the medium. m_1 and m_2 : constants that reflect the change in speed as the spatial variable changes.

This profile will be referred to as the constant velocity gradient or CVG profile, since the velocity of the ray increases linearly as the distance from r_0 increases. We can see a representation of this profile in Figure B.1.

Trajectories will be plotted for this profile by exploiting the analogy between rays and particles. An RK4 approximation will be used in order to find the trajectories of this profile.

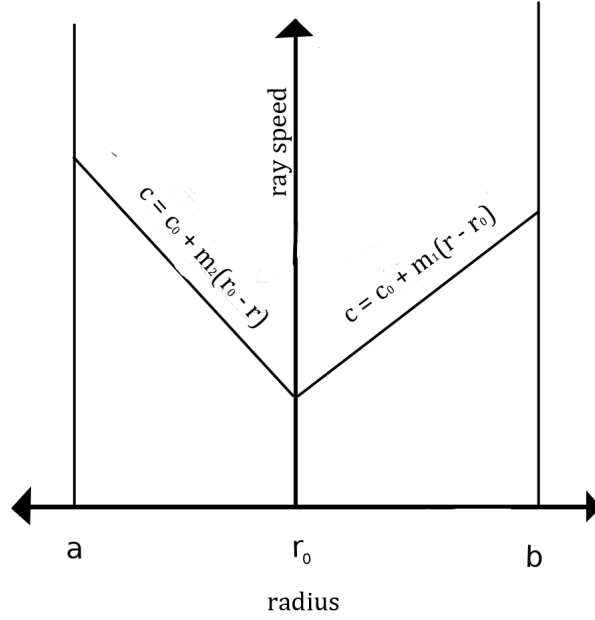


Figure B.1: A plot showing the speed of the ray vs. r

B.2 Setting up the Equations of Motion For The CVG Profile

We can determine the index of refraction profile that corresponds to the speed of sound profile above by using a relation between the refractive index and the speed of the ray.

$$\frac{n_r}{n_0} = \frac{c_0}{c_r} \quad (\text{B.2})$$

Where c_0 is the minimum speed of the ray at the medium. c_r and c_r are the speed of the ray and the refractive index at a distance z . [15]

If we rescale n_0 so that it is 1 to simplify the problem, and if we apply the speed of sound functions in (1.1), we can rewrite (2.1) as:

$$n(r) = \begin{cases} \frac{c_0}{c_0 + m_1(r - r_0)} & , b > r > r_0 \\ \frac{c_0}{c_0 + m_2(r_0 - r)} & , a < r \leq r_0 \end{cases} \quad (\text{B.3})$$

Or, if we define $k_1 = m_1/c_0$ and $k_2 = m_2/c_0$, we will have:

$$n(r) = \begin{cases} \frac{1}{1+k_1(r-r_0)} & , r_0 < r < b \\ \frac{1}{1+k_2(r_0-r)} & , a < r \leq r_0 \end{cases} \quad (\text{B.4})$$

An equation of motion describing $\frac{d^2r}{dt^2}$ in terms of r can be derived from this index of refraction profile, using the ray-particle analogy. From equation 2.2, we can find a corresponding $V(\vec{r})$ from this index of refraction:

$$V(r) = -\frac{1}{2}n^2(r) = \begin{cases} \frac{-1}{[1+k_1(r-r_0)]^2} & , r_0 < r < b \\ \frac{-1}{[1+k_2(r_0-r)]^2} & , a < r \leq r_0 \end{cases} \quad (\text{B.5})$$

We can derive these in terms of r , giving us expressions for force. With $m=1$:

$$-\frac{d}{dr}V(r) = F = \frac{d^2r}{dt^2} = \begin{cases} \frac{-k_1}{[1+k_1(r-r_0)]^3} & , r_0 < r < b \\ \frac{k_2}{[1-k_2(r_0-r)]^3} & , a < r \leq r_0 \end{cases} \quad (\text{B.6})$$

The change in angular coordinates can be calculated from the conservation of angular momentum:

$$\dot{\theta} = l/r^2 \quad (\text{B.7})$$

Where l is the angular momentum.

We will be doing a numerical approximation for this piecewise second order differential equation, and we can compare it with the orbits obtained.

The linear variation of ray speed is a commonly used model for the behaviour of sound in the ocean and atmosphere, where the index is dependent on depth or height. Used in the context of this paper, the r in these profiles is the distance from that point to the center of the earth.

B.3 Plots for the CVG Profile

Again, our CVG profile is given by:

$$n(z) = \begin{cases} \frac{1}{1+k_1(r-r_0)} & , r_0 < r < b \\ \frac{1}{1+k_2(r_0-r)} & , a < r \leq r_0 \end{cases} \quad (\text{B.8})$$

In this profile we have used real world values to see if our analysis is consistent. We will be trying to emulate ray propagation across a SOFAR channel. The following values were used:

$$\begin{aligned} m_1 = m_2 = m &= 0.02 \text{ s}^{-1} \\ c_0 &= 1500 \text{ m/s} \\ k_1 = k_2 &= m/c_0 \\ r_0 &= 6377950 \text{ m} \\ a &= 6374380 \text{ m} \\ b &= 6378100 \text{ m} \end{aligned} \quad (\text{B.9})$$

These values are based on real world values, where c_0 approximates the speed of sound in the ocean [17]. b is the average radius of the earth [15], a is the average seafloor depth of the earth, and b is the distance from the center of the earth to the seafloor [18]. Though thermocline depth varies with geography[19], a typical thermocline depth was chosen and r_0 is the distance from the center of the earth to the thermocline.[20]

Using an RK4 approximation to solve the differential equations in (2.27), trajectories were plotted in a cartesian coordinate system.

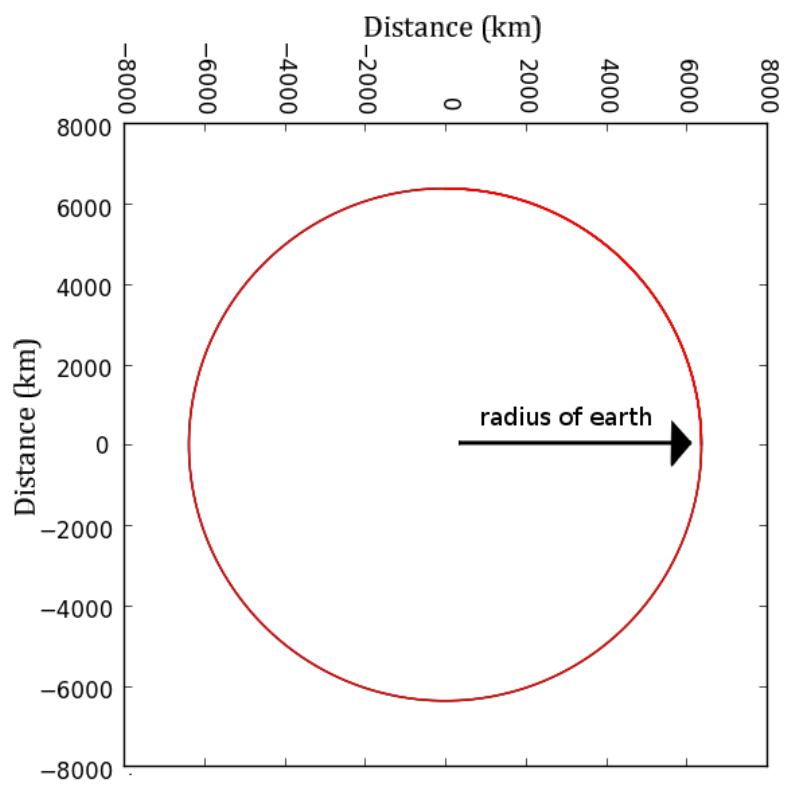


Figure B.2: Resulting plot of Python

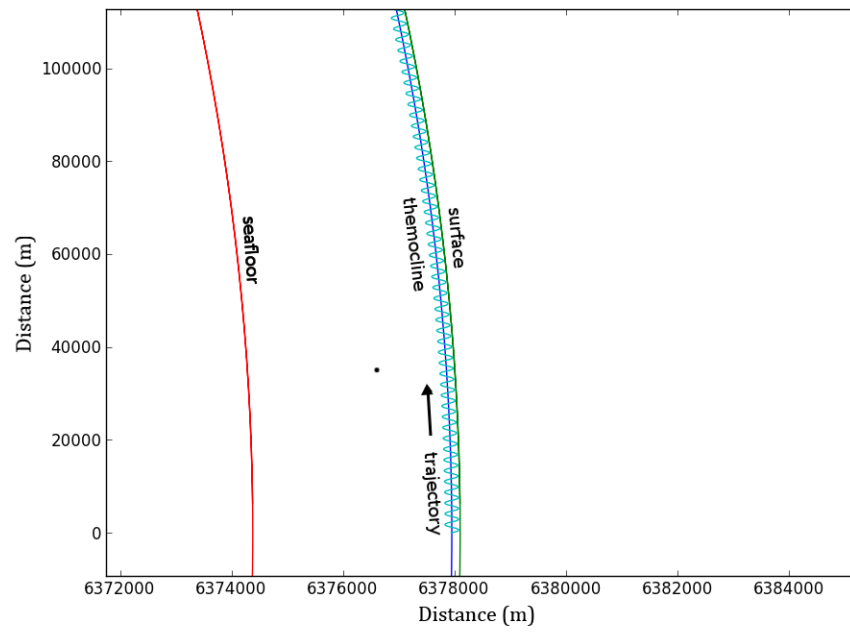


Figure B.3: Zooming in. The trajectory of the ray is the cyan line.

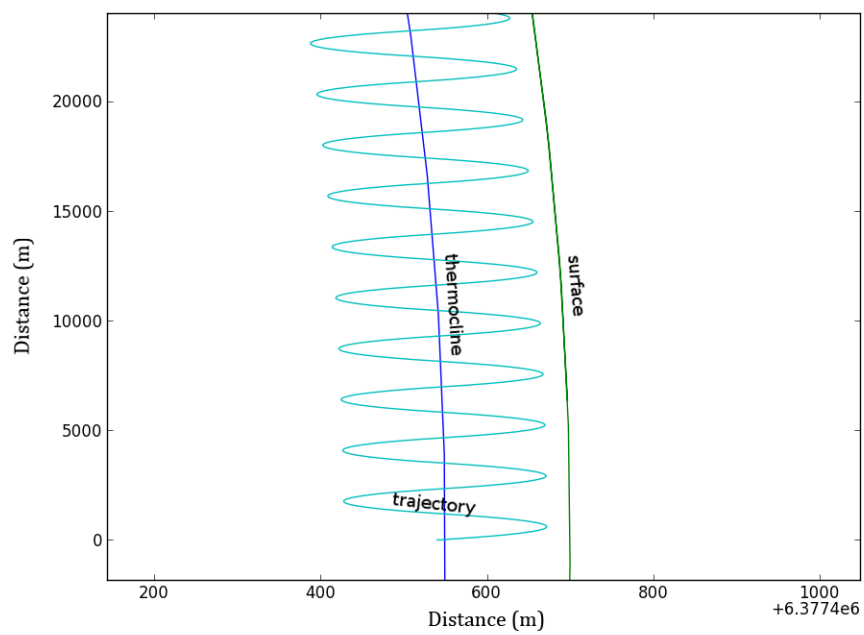


Figure B.4: Zooming in further.

We can see that the ray travels in an oscillatory motion about the thermocline, which is typical of sound propagated along SOFAR channels.

The amplitude and the wavelength of this ray were collected in Python, and these are resulting values:

$$Amplitude = 249.78 \text{ m}$$

$$Wavelength = 2.39 \text{ km}$$

These resulting values are typical of sound trajectories in shallow-water SOFAR channels. [10]

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